

TECHBIG SOLUTIONS

Ambattur, Chennai

Advanced Microsoft Excel – Complete E2E Training Syllabus

Advanced Formulas | Power Query | Power Pivot | DAX | VBA | Dashboards | Automation

Program Objective: Build advanced, job-ready Excel skills covering complex formulas, dynamic arrays, PivotTables, Power Query, Power Pivot, DAX, dashboards, financial modeling, forecasting, Solver, VBA, Office Scripts, Power Automate and real-world MIS automation.

Reference basis: Microsoft Excel learning resources covering formulas, data analysis, Power Query, Power Pivot, DAX, automation and modern Excel capabilities.

Program	Advanced Microsoft Excel – Complete E2E
Level	Intermediate → Advanced MIS / Data Analyst / Finance / Business Analyst
Core Areas	Advanced Formulas PivotTables Power Query Dashboards
Advanced Areas	Power Pivot DAX VBA Office Scripts Power Automate Modeling
Suggested Duration	100–150 hours, customizable by batch and project depth

1. Advanced Excel Overview & Professional Workflow

- Advanced Excel capabilities and business use cases
- Workbook architecture and best practices
- Data preparation → analysis → reporting → automation workflow
- Professional spreadsheet standards
- Performance and maintainability principles

2. Workbook, Worksheet & Table Management

- Advanced worksheet organization
- Excel Tables and structured references
- Named ranges and dynamic ranges
- Workbook links and dependencies
- Protection and workbook controls
- Templates and reusable models

3. Advanced Data Entry & Data Validation

- Custom validation rules
- List-based validation
- Dependent drop-downs
- Input controls and error messages
- Data-entry forms and protection
- Duplicate and invalid-data prevention

4. Advanced Formatting & Conditional Formatting

- Custom number formats
- Rule-based conditional formatting
- Formula-driven formatting

- Icon sets, data bars and color scales
- Dynamic dashboard formatting
- Print-ready professional report design

5. Advanced Formulas & Functions

- Formula evaluation and error handling
- IF, IFS, SWITCH and logical functions
- AND, OR, NOT and nested logic
- SUMIFS, COUNTIFS and AVERAGEIFS
- XLOOKUP, XMATCH and advanced lookup patterns
- INDEX/MATCH
- Dynamic-array functions and formula design

6. Dynamic Array & Modern Excel Functions

- Dynamic arrays and spill behavior
- FILTER, SORT, SORTBY and UNIQUE
- SEQUENCE and RANDARRAY
- TAKE, DROP, CHOOSECOLS and CHOOSEROWS
- VSTACK, HSTACK and TOCOL/TOROW concepts
- Modern formula-driven reporting

7. Lookup, Reference & Retrieval Techniques

- XLOOKUP advanced patterns
- Two-way lookups
- Approximate and exact matching
- INDEX/MATCH combinations
- XMATCH
- Multiple-criteria lookups
- Error-safe lookup models
- Cross-sheet and cross-workbook retrieval

8. Date & Time Intelligence

- Date/time functions
- Working-day calculations
- NETWORKDAYS and WORKDAY
- Month/quarter/year analysis
- Fiscal calendars
- Period comparisons
- YTD/MTD/QTD calculations
- Aging and SLA calculations

9. Text, Cleaning & Data Transformation Functions

- TEXT, LEFT, RIGHT, MID and extraction
- SEARCH/FIND and parsing
- TRIM, CLEAN and SUBSTITUTE

- TEXTJOIN and concatenation
- VALUE and type conversion
- Data standardization and cleansing formulas

10. Excel Tables & Structured References

- Table creation and design
- Structured-reference formulas
- Calculated columns
- Table expansion and dynamic reporting
- Table-based validation
- Combining Tables with PivotTables and Power Query

11. PivotTables – Advanced

- PivotTable creation and field layout
- Grouping dates and categories
- Calculated fields/items concepts
- Show Values As
- Running totals and percentage analysis
- Top/Bottom analysis
- Multiple PivotTables and shared caches
- PivotTable refresh management

12. PivotCharts & Interactive Reporting

- PivotCharts
- Slicers and Timelines
- Interactive filtering
- Dashboard navigation
- Dynamic chart titles
- KPI reporting
- Executive-level Pivot dashboards

13. Power Query – Data Import & Transformation

- Power Query Editor
- Importing Excel, CSV, folders and databases
- Data profiling
- Merge and Append
- Pivot/Unpivot
- Split and transform columns
- Data types and error handling
- Query parameters and reusable transformations

14. Power Query M Language

- M language concepts
- Custom columns and functions
- Parameters

- Conditional transformations
- Query dependencies
- Query folding concepts
- Reusable data-preparation workflows

15. Data Modeling & Power Pivot

- Power Pivot overview
- Data Model and relationships
- Fact and dimension tables
- Star-schema principles
- Keys and relationship design
- Calculated columns and measures
- Model performance and data-size management

16. DAX for Excel Power Pivot

- DAX fundamentals
- Measures vs calculated columns
- CALCULATE and filter context
- SUMX/AVERAGEX and iterators
- FILTER and context modification
- Time-intelligence measures
- KPIs and advanced business calculations

17. Advanced Charts & Visualization

- Combo charts
- Waterfall and funnel charts
- Scatter and bubble charts
- Histogram and Pareto analysis
- Dynamic charts
- Interactive dashboards
- Chart formatting and storytelling
- Visualization selection for business KPIs

18. Advanced Dashboard Development

- Dashboard layout and grid design
- KPI cards and scorecards
- Interactive filters and slicers
- Dynamic titles and labels
- Navigation controls
- Print/export design
- Executive dashboard standards

19. What-If Analysis & Decision Models

- Goal Seek
- Scenario Manager

- Data Tables
- What-If Parameters
- Sensitivity analysis
- Break-even analysis
- Scenario comparison and decision support

20. Financial & Business Modeling

- Revenue and expense models
- Budget vs actual
- Forecast models
- Cash-flow modeling
- Profitability analysis
- Loan/EMI calculations
- Investment and ROI models
- Scenario and sensitivity models

21. Forecasting & Statistical Analysis

- Forecast Sheet
- Trend and seasonality concepts
- Moving averages
- Descriptive statistics
- Correlation and regression concepts
- Variance and standard deviation
- Business forecasting models

22. Solver & Optimization

- Solver setup
- Objective cells and decision variables
- Constraints
- Linear and nonlinear optimization concepts
- Resource allocation
- Scheduling and optimization scenarios

23. Data Analysis ToolPak

- Descriptive statistics
- Histogram
- Correlation
- Regression
- Moving average
- Sampling
- ANOVA overview
- Business interpretation of statistical output

24. Advanced Filtering, Sorting & Data Tools

- Advanced Filter

- Multi-level sorting
- Custom sorting
- Remove duplicates
- Text to Columns
- Flash Fill
- Subtotal and outline
- Data consolidation

25. Import/Export & External Data

- CSV and text data
- Folder-based data
- SQL Server connectivity concepts
- Web/external data concepts
- Refreshable data connections
- External workbook links
- Data source governance

26. Excel Automation with VBA

- VBA environment and macro security
- Macro recording and editing
- Variables, data types and control structures
- Loops and conditions
- Procedures and functions
- Workbook/worksheet/range objects
- Events and automation patterns
- Error handling and debugging

27. Advanced VBA & Excel Automation

- Dynamic VBA ranges
- Arrays and dictionaries
- UserForms
- File and folder automation
- Automated report generation
- Email/report distribution concepts
- Performance optimization
- Reusable VBA modules and maintainable automation

28. Office Scripts & Modern Automation

- Office Scripts overview
- Automating repetitive Excel tasks
- TypeScript-based scripting concepts
- Cloud workbook automation
- Power Automate integration concepts
- Choosing VBA vs Office Scripts

29. Power Automate & Excel Integration

- Excel with Power Automate
- Automated approvals and notifications
- Scheduled data workflows
- Excel Online tables
- File-based automation
- Business-process automation scenarios
- Error handling and monitoring

30. Excel Security & Governance

- Worksheet/workbook protection
- Password and access concepts
- Macro security
- Safe external links
- Data privacy
- Sensitive-data handling
- Version control and audit considerations

31. Performance Optimization

- Workbook calculation modes
- Formula optimization
- Avoiding volatile functions
- Efficient lookup and array formulas
- Large-data strategies
- Pivot/Power Query optimization
- VBA performance techniques
- Workbook diagnostics

32. Printing, Presentation & Professional Reporting

- Page setup and print areas
- Headers and footers
- Page breaks
- Repeat rows/columns
- PDF-ready reporting
- Executive report formatting
- Professional templates and standards

33. Real-Time E2E Projects

- Sales MIS dashboard with automated monthly reporting
- Finance budget-vs-actual and P&L; dashboard
- HR attendance, payroll and employee analytics workbook
- Inventory aging, stock movement and reorder dashboard
- Procurement spend and supplier-performance analysis
- Production KPI and downtime analysis

- Advanced MIS using Power Query + Pivot + dashboard
- Financial forecasting and scenario model with Solver
- Automated Excel report generation using VBA/Office Scripts
- End-to-end Excel reporting system with Power Query, Power Pivot, DAX, automation and governance

34. Troubleshooting & Production Support

- Formula errors and circular references
- Broken links and external-reference issues
- Pivot refresh failures
- Power Query refresh errors
- Data-model relationship issues
- VBA runtime errors
- Macro/security problems
- Slow workbook diagnosis
- Data-quality issues
- Root-cause analysis and support practices

35. Interview & Practical Assessment

- Advanced Excel interview questions
- Formula and lookup case studies
- PivotTable and dashboard exercises
- Power Query practical scenarios
- Power Pivot/DAX questions
- VBA automation case studies
- Business-modeling assignments
- Real-time project explanation
- Mock practical assessment

Recommended Hands-on Deliverables

- Build an automated MIS dashboard using Excel Tables, Power Query, PivotTables, slicers and advanced formulas.
- Create a Power Pivot data model with relationships, DAX measures and executive KPIs.
- Develop an advanced financial model with scenarios, forecasting, sensitivity analysis and Solver.
- Automate a recurring business report using VBA or Office Scripts and integrate notification/distribution workflows.
- Design a large-data workbook and optimize formulas, queries, pivots and calculation performance.
- Complete a production-style Excel reporting solution with validation, protection, documentation and support procedures.

Training Outcome

Learners completing this program should be able to independently analyze and transform business data, build advanced Excel models and dashboards, use Power Query/Power Pivot/DAX, automate repetitive processes, create professional MIS and financial reports, and troubleshoot production Excel solutions.